

TFG - 5. Anàlisi de bombolles especulatives

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Lectura de les dades

```
dades_netes <- readRDS("C:/Users/joelm/Downloads/TFG/dades_netes.rds")
summary(dades_netes)
```

```
##      ID_Card      Date      Price      Vintage
## Length:20559   Min.   :2016-09-12   Min.   : 10.0   Moderna:10887
## Class :character 1st Qu.:2021-01-02   1st Qu.: 152.5   Vintage: 9672
## Mode  :character Median :2022-09-13   Median : 375.0
##                      Mean  :2022-05-18   Mean   : 595.0
##                      3rd Qu.:2024-04-11   3rd Qu.: 800.0
##                      Max.   :2026-03-02   Max.   :9700.0
```

```
llista_cartes <- unique(dades_netes$ID_Card)
if(!dir.exists("grafics_bombolles2")) dir.create("grafics_bombolles2")
```

```
library(dplyr)
library(lubridate)
```

```
#install.packages("remotes")
#remotes::install_github("itamarcaspi/psymonitor")
```

```
library(psymonitor)
library(zoo)
library(ggplot2)
library(tidyr)
```

Creació del bucle i implementació mètode PSY

```
set.seed(123)

for (carta_id in llista_cartes) {

  print(paste("Processant:", carta_id))

  # Preparació de dades
  df_temp <- dades_netes %>%
```

```

filter(ID_Card == carta_id) %>%
mutate(Mes_Any = floor_date(Date, "month")) %>%
group_by(Mes_Any) %>%
summarise(Preu_Mensual = mean(Price, na.rm = TRUE), .groups = 'drop') %>%
complete(Mes_Any = seq(min(Mes_Any), max(Mes_Any), by = "month")) %>%
mutate(Preu_Mensual = na.locf(Preu_Mensual, na.rm = FALSE)) %>%
mutate(Preu_Mensual = na.locf(Preu_Mensual, fromLast = TRUE))

obs_totals <- nrow(df_temp)

# micro-soroll blanc per evitar variància zero
soroll_blanc <- rnorm(obs_totals, mean = 0, sd = 0.0001)
y <- as.numeric(df_temp$Preu_Mensual) + soroll_blanc
dates <- df_temp$Mes_Any

# Paràmetres PSY
r0 <- 0.01 + 1.8 / sqrt(obs_totals)
swindow0 <- floor(r0 * obs_totals)
dim <- obs_totals - swindow0 + 1
IC <- 2

adflag <- 1

Tb <- min(12 * 2 + swindow0 - 1, obs_totals)

# Execució del Test PSY
bsadf <- tryCatch({
  PSY(y, swindow0, IC, adflag)
}, error = function(e) {
  print(paste("Error matemàtic intern a", carta_id, "tot i les correccions. Saltant."))
  return(NULL)
})

if(is.null(bsadf)) next

quantilesBsadf <- cvPSYwmboot(y, swindow0, IC, adflag, Tb, nboot = 999, nCores = 2)

# Processament de resultats
dates_test <- as.Date(dates[swindow0:obs_totals])
quantiles_matrix <- quantilesBsadf %>% matrix(1, nrow = 1, ncol = dim)

ind95 <- as.numeric((bsadf > t(quantiles_matrix[2, ])) * 1)

df_plot <- data.frame(
  Date = dates_test,
  Price = as.numeric(y[swindow0:obs_totals]),
  Bubble = ind95
)

df_bombolles <- df_plot %>% filter(Bubble == 1)

# Generació del gràfic
p <- ggplot(df_plot, aes(x = Date, y = Price)) +

```

```

geom_rect(data = df_bombolles,
          aes(xmin = Date, xmax = Date + months(1), ymin = -Inf, ymax = Inf),
          fill = "forestgreen", alpha = 0.45, inherit.aes = FALSE) +
geom_line(linewidth = 1) +
theme_minimal(base_size=18) +

theme(
  plot.title = element_text(size = 25, face = "bold", hjust = 0.5, margin = margin(b = 15)),
  axis.title = element_text(size = 28),
  axis.text = element_text(size = 22),
  plot.margin = margin(20, 20, 20, 20),
  plot.background = element_rect(fill = "white", color = NA),
  panel.background = element_rect(fill = "white", color = NA)
) +

labs(title = carta_id, x = "Data", y = "Preu (USD)")

ggsave(filename = paste0("graphics_bombolles95/", carta_id, ".png"), plot = p, width = 10, height = 6)
}

```

```

## [1] "Processant: vin1_charizard_holo_baseset_psa9"

## [1] "Processant: vin2_blastoise_holo_baseset_psa9"

## [1] "Processant: vin3_venusaur_holo_baseset_psa9"

## [1] "Processant: vin4_mewtwo_holo_baseset_psa9"

## [1] "Processant: vin5_pikachu_yellowcheeks_baseset_psa9"

## [1] "Processant: vin6_dragonite_holo_fossil_psa9"

## [1] "Processant: vin7_gengar_holo_fossil_psa9"

## [1] "Processant: vin8_darkblastoise_holo_teamrocket_psa9"

## [1] "Processant: vin9_darkcharizard_holo_teamrocket_psa9"

## [1] "Processant: vin10_ancientmew_promomovie_psa9"

## [1] "Processant: mod1_mewtwo_gx_sr_shininglegends_psa10"

## [1] "Processant: mod2_latiasandlatiosgx_altart_teamup_psa10"

## [1] "Processant: mod3_charizardgx_shiny_hiddenfates_psa10"

## [1] "Processant: mod4_charizardv_shiny_championspath_psa10"

## [1] "Processant: mod5_pikachuvmax_rainbow_vividvoltage_psa10"

```

```
## [1] "Processant: mod6_charizardvmax_shiny_shiningfates_psa10"

## [1] "Processant: mod7_sylveonvmax_altart_evolving skies_psa10"

## [1] "Processant: mod8_umbreonvmax_altart_evolving skies_psa10"

## [1] "Processant: mod9_rayquazavmax_altart_evolving skies_psa10"

## [1] "Processant: mod10_gengarvmax_altart_fusionstrike_psa10"
```

```
print("Procés finalitzat. Revisa la carpeta 'grafics_bombolles'.")
```

```
## [1] "Procés finalitzat. Revisa la carpeta 'grafics_bombolles'."
```

Grafiquem tot junt

```
library(patchwork)
library(png)
library(grid)

directori_origen <- "grafics_bombolles95/"
directori_desti <- "./"

carregar_imatge_com_plot <- function(nom_fitxer) {
  img <- readPNG(paste0(directori_origen, nom_fitxer))
  g <- rasterGrob(img, interpolate = TRUE)

  p <- ggplot() +
    annotation_custom(g, xmin=-Inf, xmax=Inf, ymin=-Inf, ymax=Inf) +
    theme_void() +
    theme(
      plot.margin = margin(5, 5, 5, 5),
      plot.background = element_rect(fill = "white", color = NA),
      panel.background = element_rect(fill = "white", color = NA)
    )
  return(p)}

# vintage

print("Generant la composició Vintage...")
```

```
## [1] "Generant la composició Vintage..."
```

```
plots_vintage <- list(
  carregar_imatge_com_plot("vin1_charizard_holo_baseset_psa9.png"),
  carregar_imatge_com_plot("vin2_blastoise_holo_baseset_psa9.png"),
  carregar_imatge_com_plot("vin3_venusaur_holo_baseset_psa9.png"),
  carregar_imatge_com_plot("vin4_mewtwo_holo_baseset_psa9.png"),
  carregar_imatge_com_plot("vin5_pikachu_yellowcheeks_baseset_psa9.png"),
```

```

    carregar_imatge_com_plot("vin6_dragonite_holo_fossil_psa9.png"),
    carregar_imatge_com_plot("vin7_gengar_holo_fossil_psa9.png"),
    carregar_imatge_com_plot("vin8_darkblastoise_holo_teamrocket_psa9.png"),
    carregar_imatge_com_plot("vin9_darkcharizard_holo_teamrocket_psa9.png"),
    carregar_imatge_com_plot("vin10_ancientmew_promomovie_psa9.png")
  )

composicio_vintage <- wrap_plots(plots_vintage, ncol = 2, nrow = 5) +
  plot_annotation(
    title = "Comparativa de Bombolles Especulatives: Cartes Vintage (95% Confiança)",
    subtitle = "Mètode PSY (Phillips, Shi i Yu, 2015). Zones verdes indiquen comportament explosiu.",
    theme = theme(
      plot.background = element_rect(fill = "white", color = NA),
      text = element_text(family = "sans"),
      plot.title = element_text(size = 25, face = "bold", hjust = 0.5, margin = margin(b = 20)),
      plot.subtitle = element_text(size = 20, hjust = 0.5, margin = margin(b = 30))
    )
  )

ggsave(paste0(directori_desti, "Vintage_Bombolles_Consolidat95.png"),
  plot = composicio_vintage, width = 14, height = 18, dpi = 300, bg = "white")

# modernes

print("Generant la composició Moderna...")

## [1] "Generant la composició Moderna..."

plots_modernes <- list(
  carregar_imatge_com_plot("mod1_mewtwox_sr_shininglegends_psa10.png"),
  carregar_imatge_com_plot("mod2_latiasandlatiosgx_altart_teamup_psa10.png"),
  carregar_imatge_com_plot("mod3_charizardgx_shiny_hiddenfates_psa10.png"),
  carregar_imatge_com_plot("mod4_charizardv_shiny_championspath_psa10.png"),
  carregar_imatge_com_plot("mod5_pikachuvmax_rainbow_vividvoltage_psa10.png"),
  carregar_imatge_com_plot("mod6_charizardvmax_shiny_shiningfates_psa10.png"),
  carregar_imatge_com_plot("mod7_sylveonvmax_altart_evolving skies_psa10.png"),
  carregar_imatge_com_plot("mod8_umbreonvmax_altart_evolving skies_psa10.png"),
  carregar_imatge_com_plot("mod9_rayquazavmax_altart_evolving skies_psa10.png"),
  carregar_imatge_com_plot("mod10_gengarvmax_altart_fusionstrike_psa10.png")
)

composicio_modernes <- wrap_plots(plots_modernes, ncol = 2, nrow = 5) +
  plot_annotation(
    title = "Comparativa de Bombolles Especulatives: Cartes Modernes (95% Confiança)",
    subtitle = "Mètode PSY (Phillips, Shi i Yu, 2015). Zones verdes indiquen comportament explosiu.",
    theme = theme(
      plot.background = element_rect(fill = "white", color = NA),
      text = element_text(family = "sans"),
      plot.title = element_text(size = 25, face = "bold", hjust = 0.5, margin = margin(b = 20)),
      plot.subtitle = element_text(size = 20, hjust = 0.5, margin = margin(b = 30))
    )
  )

```

```
ggsave(paste0(directori_desti, "Modernes_Bombolles_Consolidat95.png"),
       plot = composicio_modernes, width = 14, height = 18, dpi = 300, bg = "white")

print("Procés finalitzat")
```

```
## [1] "Procés finalitzat"
```